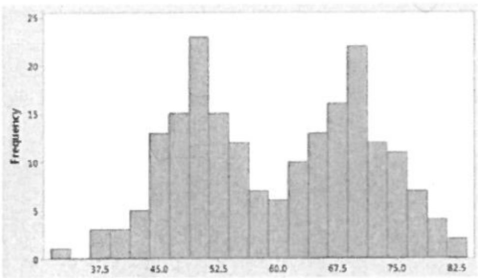
Name:	Answers - Version 2	Date:________
Teacher:	________________	
Baldivis Secondary College	**Year 9 Mathematics** **Mini test 4** **Topic – Data Analysis** ***Full working out MUST be shown to get full marks for each question.***	SCORE: / 28
Total Time:	30 minutes	
Weighting:	5 %	
Equipment:	To be provided by the student: Pens and Pencils, scientific calculator, A4 page of notes.	

1) Underneath each graph below, write the most appropriate term to describe the distribution.

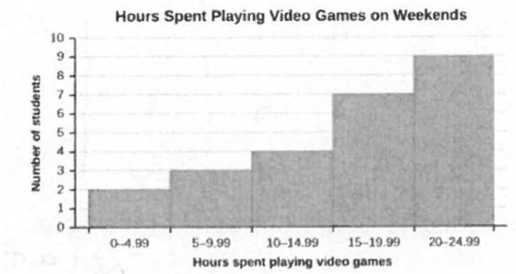
[4 marks]

a)



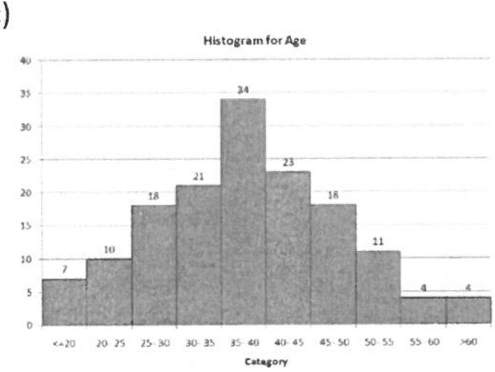
Bimodal

b)



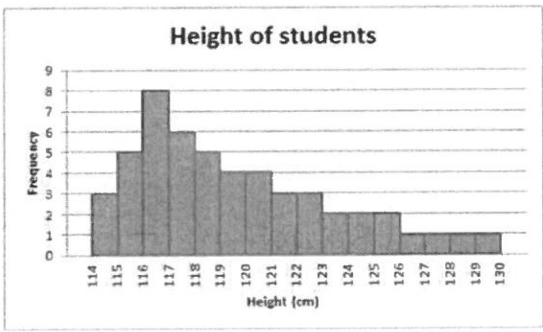
Negative skew

c)



Symmetrical

d)



Positive skew

2. Isagani and Lara were arguing whose basketball team was better. They looked at the scores from their games last year:

[1,3,1,2,2,3 - 12 marks]

Isagani's team	55, 78, 84, 98, 77, 86, 88, 101, 95, 32, 65, 63, 75, 81, 59
Lara's team	47, 34, 38, 100, 54, 51, 43, 42, 47, 58, 65, 57, 85, 91, 63

a) Is the above data continuous or discrete?

discrete

b) Display the above data using the appropriate graphical display

Isagani		Lara
2	3	4 8
	4	2 3 7 7
9 5	5	1 4 7 8
5 3	6	3 5
8 7 5	7	
8 6 4 1	8	5
8 5	9	1
1	10	0

$\boxed{\text{key} = 3|4 = 34}$

✓

✓ - ordered

✓ - key

c) Describe the shape of both team's data.

Isagani - ~~positive~~ negative skew

Lara - positive skew ✓ - 1mk

d) Calculate each team's:

i) Median

Isagani - 78 ✓

Lara - 54 ✓

ii) Range

Isagani - 101 - 32 = 69 ✓

Lara - 100 - 34 = 66 ✓

iii) Decide which team performed better last year? Justify your answer.

Isagani - ✓ 1 mk

Any two reasonable justifications - ✓✓ 2mks

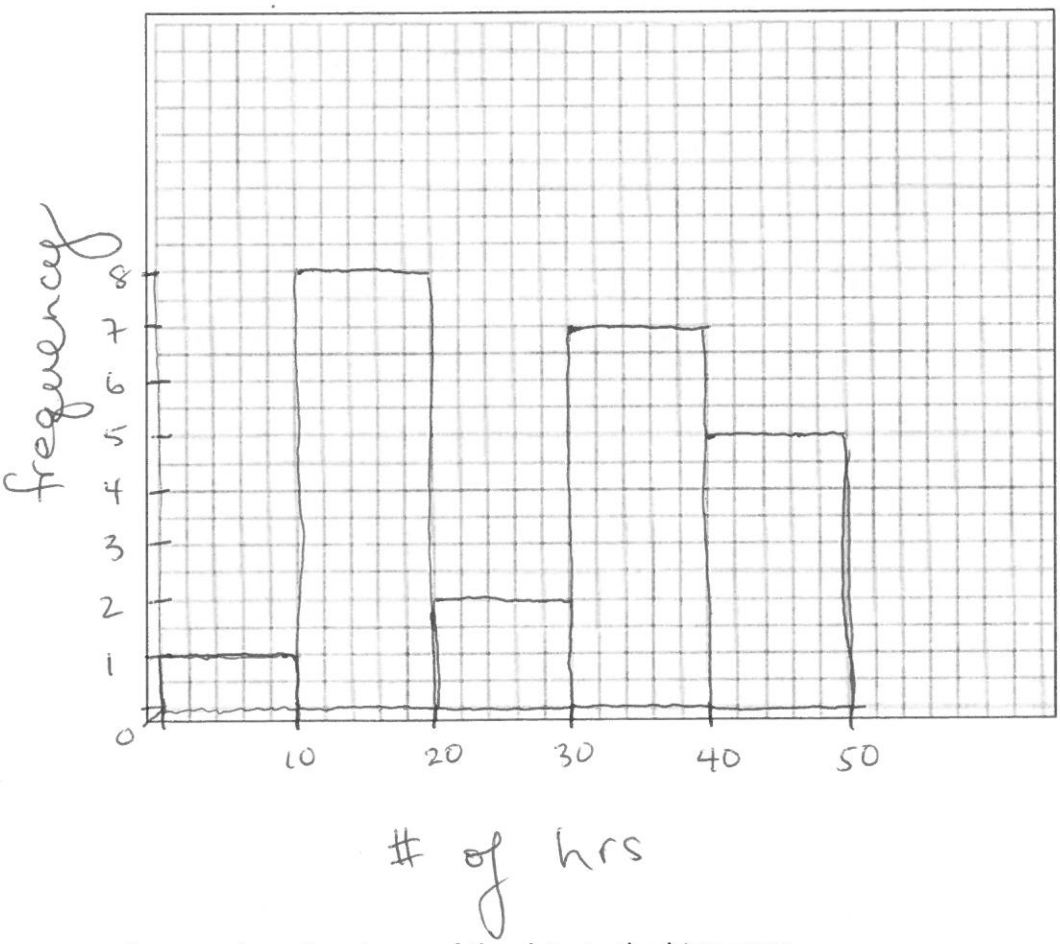
3) The data below is information about the number of hours worked by different employees during 1 week at Built Fast Enterprises. [3 + 4 + 2 + 3 = 12 marks]

Week 1 data: 14, 17, 6, 42, 18, 13, 16, 17, 40, 19, 39, 31, 37, 45, 30, 21, 22, 38, 39, 36, 48, 49, 18

a. create a frequency table for this data.

0 - <10	1
10 - <20	8
20 - <30	2
30 - <40	~~8a~~ 7
40 - <50	5

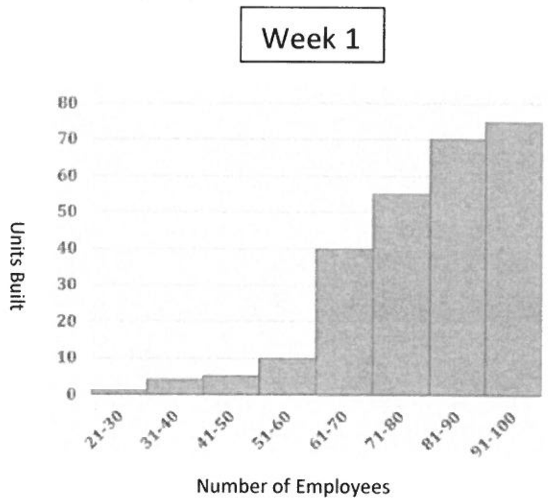
b. Use the frequency table to create a histogram.

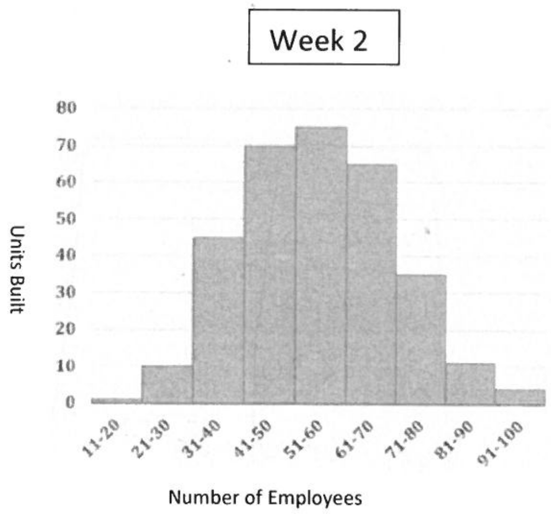


c. Comment on the shape of the data in the histogram.

Follow through based on histogram

d. Later, Built Fast analysed two separate weeks to look at the productivity of their employees during their working hours. They compiled their results below:





Comment on the distribution of both data sets and explain which week was more productive, using specific reference to how each week's distribution impacts the measures of centre.

✓ 1 mk - wk 1 was more productive

✓✓ 2mks - Any 2 reasonable justifications, given they aren't repetitive.